

CS509 Laboratory Repository

Repository Overview

This repository contains buddy (pair) assignments for CS509 - Parallel Computer Architecture. Assignment 01 implements BFS, DFS, and SSSP (Dijkstra) on graphs stored in CSR (Compressed Sparse Row) format.

Student / Pair Details

- **Student 1:** Vishal Chandel (2026AIM1023)
- **Student 2:** (Buddy partner entry number)
- **Program:** M.Tech Artificial Intelligence, IIT Ropar

Language and Environment

- **Language:** C++17
- **Compiler:** MinGW g++ (GCC)
- **Build Tool:** mingw32-make
- **Timing Method:** `std::chrono::high_resolution_clock`, reported in milliseconds (ms)
- **Platform:** Windows 10/11

Directory Structure

```
CS_509_2026AIM1023_2026AIM1023/  
|-- README.md  
|-- common_wrapper/  
|   |-- wrapper.cpp  
|   |-- Makefile  
|-- assignment_01/  
|   |-- driver/  
|   |   |-- driver.cpp  
|   |-- include/  
|   |   |-- bfs.h  
|   |   |-- csr.h  
|   |   |-- dfs.h  
|   |   |-- graph.h
```

```

| |-- sssp.h
| |-- timer.h
|-- src/
| |-- bfs.cpp
| |-- csr.cpp
| |-- dfs.cpp
| |-- graph.cpp
| |-- sssp.cpp
| |-- timer.cpp
|-- tests/
| |-- bfs_10.txt
| |-- bfs_100.txt
| |-- bfs_10000.txt
| |-- bfs_50000.txt
| |-- bfs_100000.txt
| |-- dfs_10.txt
| |-- dfs_100.txt
| |-- dfs_10000.txt
| |-- dfs_50000.txt
| |-- dfs_100000.txt
| |-- sssp_10.txt
| |-- sssp_100.txt
| |-- sssp_10000.txt
| |-- sssp_50000.txt
| |-- sssp_100000.txt
|-- Makefile

```

Common Wrapper: Build and Usage

The wrapper acts as a main program interface.

```

cd common_wrapper
mingw32-make
.\wrapper.exe

```

The wrapper provides:

- Assignment selection menu
- Compile Assignment 01
- Run one selected test file
- Run all tests for one algorithm
- Run all algorithms on all test files

General Conventions

- Graph input uses adjacency-list format with a `SOURCE s` configuration line.

- Adjacency-list-to-CSR conversion is done during preprocessing and is excluded from timing metrics.
- Timing includes only algorithm execution time.
- Vertex numbering uses 0 to V-1.
- Variables use type-prefixed snake_case naming style.

Assignment 01 - BFS, DFS, and SSSP

Assignment Mode

Double / Buddy

Objective

Implement graph traversal (BFS, DFS) and shortest-path (SSSP) algorithms using CSR representation.

Algorithm / Approach

- **CSR Conversion:** Converts adjacency representation to flat `v_row_ptr`, `v_col_index`, and `v_weights` vectors.
- **BFS:** Queue-based level-order traversal with unweighted edge-count distance calculations.
- **DFS:** Iterative depth-first traversal. Uses a heap-allocated `std::vector` stack instead of deep call recursion frames to avoid Windows thread stack limit constraints (stack overflows on V=100,000 scale).
- **SSSP:** Dijkstra's shortest-path algorithm using a binary min-priority queue (positive weights only).

Input Format

Unweighted (BFS/DFS):

```
V E
u degree neighbor1 neighbor2 ...
...
SOURCE s
```

Weighted (SSSP):

```
V E
u degree neighbor1 weight1 neighbor2 weight2 ...
...
SOURCE s
```

Helper Functions / CSR Conversion

- `read_unweighted_graph()` / `read_weighted_graph()` in `src/graph.cpp`
- `convert_to_csr()` in `src/csr.cpp`
- Preprocessing time is excluded from reported algorithm runtimes.

File Structure

File	Purpose
-----	-----
<code>driver/driver.cpp</code>	Command-line parsing, CSR conversion preprocessing, algorithm dispatch, formatted console logging.
<code>src/graph.cpp</code>	Adjacency list graph reader helper.
<code>src/csr.cpp</code>	Conversion utilities to CSR format.
<code>src/bfs.cpp</code>	BFS level-order queue search.
<code>src/dfs.cpp</code>	DFS iterative stack traversal.
<code>src/sssp.cpp</code>	Dijkstra min-priority queue shortest path search.

Compilation

```
cd assignment_01
mingw32-make
```

Execution

```
# Run a specific algorithm on a test file
.\graph_simulator.exe bfs tests/bfs_10.txt
```

```

.\graph_simulator.exe dfs tests/dfs_10.txt
.\graph_simulator.exe sssp tests/sssp_10.txt

# Run all tests for a specific algorithm
.\graph_simulator.exe --all-tests bfs

# Run all algorithms on all scales (10, 100, 10000, 50000, 100000)
.\graph_simulator.exe --all

```

Test Cases and Result Table

All benchmarks were run locally on the target platform.

Algorithm	Test File	Vertices	Edges	Input Type	Source	Expected Output	Actual Output	Time (ms)	Status
-----	-----	----- --	-----	-----	----- -	----- -----	----- -----	-----	----- --
BFS	bfs_10.txt	10	14	Unweighted adjacency list (CSR)	0	Traversal + distances	Matches expected	0.000 ms	Pass
BFS	bfs_100.txt	100	200	Unweighted adjacency list (CSR)	0	Traversal + distances	Matches expected	0.000 ms	Pass
BFS	bfs_10000.txt	10,000	20,000	Unweighted adjacency list (CSR)	0	Traversal + distances	Matches expected	1.994 ms	Pass
BFS	bfs_50000.txt	50,000	100,000	Unweighted adjacency list (CSR)	0	Traversal + distances	Matches expected	13.832 ms	Pass
BFS	bfs_100000.txt	100,000	200,000	Unweighted adjacency list (CSR)	0	Traversal + distances	Matches expected	21.210 ms	Pass
DFS	dfs_10.txt	10	14	Unweighted adjacency list (CSR)	0	Valid DFS traversal	Matches expected	0.000 ms	Pass
DFS	dfs_100.txt	100	200	Unweighted adjacency list (CSR)	0	Valid DFS traversal	Matches expected	0.000 ms	Pass
DFS	dfs_10000.txt	10,000	20,000	Unweighted adjacency list (CSR)	0	Valid DFS traversal	Matches expected	3.007 ms	Pass

Algorithm	Test File	Vertices	Edges	Input Type	Source	Expected Output	Actual Output	Time (ms)	Status
DFS	dfs_50000.txt	50,000	100,000	Unweighted adjacency list (CSR)	0	Valid DFS traversal	Matches expected	15.383 ms	Pass
DFS	dfs_100000.txt	100,000	200,000	Unweighted adjacency list (CSR)	0	Valid DFS traversal	Matches expected	91.566 ms	Pass
SSSP	sssp_10.txt	10	16	Positive weighted adjacency list (CSR)	0	Shortest distances	Matches expected	0.000 ms	Pass
SSSP	sssp_100.txt	100	200	Positive weighted adjacency list (CSR)	0	Shortest distances	Matches expected	0.000 ms	Pass
SSSP	sssp_10000.txt	10,000	20,000	Positive weighted adjacency list (CSR)	0	Shortest distances	Matches expected	23.589 ms	Pass
SSSP	sssp_50000.txt	50,000	100,000	Positive weighted adjacency list (CSR)	0	Shortest distances	Matches expected	136.978 ms	Pass
SSSP	sssp_100000.txt	100,000	200,000	Positive weighted adjacency list (CSR)	0	Shortest distances	Matches expected	319.434 ms	Pass

Complexity

- **CSR Conversion:** Time complexity $O(V + E)$, Space complexity $O(V + E)$
- **BFS / DFS:** Time complexity $O(V + E)$, Space complexity $O(V + E)$
- **Dijkstra (SSSP):** Time complexity $O((V + E) \log V)$ with binary heap, Space complexity $O(V + E)$

References

- CS509 Assignment 1 Specification
- CS509 Lab Work Guidelines